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Exposé — Bachelor's Thesis

Techniques for using NAS in FL

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1 Problem Statement

Deep learning has seen widespread adoption over the last decade and is being applied to an increasingly diverse set of tasks. Applying deep learning traditionally involves a team of deep learning and domain experts who tailor a neural network architecture to a specific task through a lengthy process of trial and error. To automate this process, researchers invented *Neural Architecture Search* (NAS) [2] methods. NAS methods employ diverse strategies to automatically search for a neural network architecture for a given deep learning task within an architecture search space.

Independently, but in parallel to NAS, researchers developed a machine learning approach called *Federated Learning* (FL) [6], which enables training an ML model on data distributed across a set of clients without sharing their data. The model is distributed from the server to the clients, which train the model on their local data. The clients then send the model weight updates to the server, which aggregates them and redistributes them to the clients. This process is repeated for several rounds. FL enhances the privacy of client data and enables training even when clients' data is too large to transfer effectively to a central training site or cannot be shared due to regulatory or privacy constraints.

When practitioners¹ make use of deep learning in FL, they typically use a predefined architecture. However, clients' data distributions and hardware capabilities are highly heterogeneous in FL. Since practitioners cannot directly observe these client characteristics [4], it is hard to choose a single architecture that generalises well across all clients and that meets inference latency constraints on every client. Using NAS methods in FL provides an alternative: the server can coordinate an architecture search across clients, guided by performance feedback from candidate architectures evaluated directly on clients. Thus, an architecture can be found that is tailored to the clients' characteristics.

Despite the benefit, using NAS in FL is not straightforward. Most NAS methods were developed for a centralised setting and rely on assumptions that often do not hold for FL. These assumption discrepancies create several *challenges* for using NAS in FL. For example, in a centralised setting, practitioners can expect a NAS method to run on worker nodes connected via low-latency, high-bandwidth links. However, in FL, a NAS method may run on clients connected via an unreliable, high-latency, low-bandwidth network. This creates a challenge when transferring the weights of large candidate models between clients and the server to coordinate an architecture search. Practitioners need specialised techniques to overcome such challenges, such as only transferring encodings of candidate architectures to clients [7].

Many techniques for overcoming such challenges already exist, but practitioners

¹In this thesis, *practitioners* refers to both users and developers of NAS methods for FL.

need to comb an increasingly large body of NAS-in-FL literature for suitable techniques (i.e., techniques that overcome the challenges relevant to their targeted FL system characteristics). The relevance of a challenge depends on the extent to which the targeted FL system characteristics violate centralised NAS assumptions. FL system characteristics include the average hardware capabilities of clients, the average network latency of clients, the number of participating clients, etc. Additionally, practitioners need to weigh which techniques to use to overcome their relevant set of challenges, since a technique for overcoming one challenge may inadvertently make it harder to overcome another.

However, finding suitable techniques is difficult. Prior surveys of the NAS-in-FL literature categorise NAS-in-FL methods as a whole rather than analyse the individual techniques they use. [8] categorises NAS-in-FL methods into offline versus online architecture search and single-objective versus multi-objective methods. [5] gives a high-level overview of the NAS-in-FL landscape at the time of its publication as part of a larger survey into combining NAS and Hyperparameter Optimisation. [3] investigates how multi-objective optimisation can be integrated into FL in general and includes only parts that discuss how this is done specifically for NAS-in-FL methods. Moreover, all of the surveys analyse only a small fraction of the entire NAS-in-FL literature. As a result, knowledge on challenges when using NAS in FL and the techniques for overcoming those challenges remains fragmented across several studies. A more sophisticated understanding of the challenges and remedying techniques for using NAS in FL is needed to allow practitioners to find techniques that could overcome the set of challenges relevant to their targeted set of FL system characteristics. Therefore, we ask the following research question:

What challenges arise when using NAS in FL, and what techniques overcome these challenges in the literature?

To tackle our research question, we conduct an extensive systematic literature review. We collect an initial set of relevant literature with a Scopus [1] search and extend it with a snowball search. We first use open coding to identify, from the literature, assumption discrepancies and resulting challenges, when using NAS in FL. Next, we extract FL system characteristics and use axial coding to code the influence of characteristics on the relevance of challenges. Then, we extract techniques through open coding and iteratively refine them to synthesise a coherent set of mutually exclusive and collectively exhaustive techniques. Finally, we use axial coding to map techniques onto challenges they overcome.

Our review aims to synthesise techniques for using NAS in FL from the literature to make finding suitable techniques easy in practice. We make the following three

contributions. First, by providing a consolidated overview of the challenges of using NAS in FL, practitioners can grasp which issues they can expect to encounter at a glance. Second, by describing how FL system characteristics influence the relevance of challenges, we enable practitioners to focus on the challenges relevant to them. Third, by extracting techniques for using NAS in FL and highlighting the challenges they address, we enable practitioners to compare existing techniques for using NAS in FL for their specific set of challenges.

2 Objectives

The primary objective of this thesis is to provide an overview of techniques for using NAS in FL, spawning the following sub-objectives:

1. Catalogue the assumptions that hold for centralised NAS, but are violated when using NAS in FL.
2. Describe the challenges arising from the mismatch in assumptions between centralised NAS and NAS in FL.
3. Describe the influence of FL system characteristics on the relevance of challenges.
4. Catalogue the techniques used to overcome these challenges.

3 Explanation of Terms

3.1 Neural Architecture Search

Neural Architecture Search [9] (NAS) enables the automatic search for neural network architecture as opposed to manual experimentation. A NAS method consists of a *search space* for potential candidate architectures, a *search strategy* that guides the architecture search, and a *performance estimation strategy* used by the search strategy to decide which candidates to favour or explore next [2].

3.2 Federated Learning

Federated Learning [6] (FL) is a machine learning approach in which multiple *clients* collaboratively train a shared model while keeping their training data local. A central *server* coordinates *communication rounds* with the clients, wherein each client trains the model for several epochs locally and finally sends gradient updates to the server for aggregation into the shared model. After aggregation, the server sends the updates to clients for the next round of training.

4 Research Approach

4.1 Data Collection

We obtain the set of literature relevant to our review by searching the abstract, title and keywords of English literature in Scopus [1] with the search query "*federated learning*" AND "*neural architecture search*" and include literature that describes a method for using NAS in FL. We do not include surveys as part of our review. We extend this initial set of literature by a single iteration of a snowball search. We exclude literature that does not provide sufficient methodological detail to extract challenges or techniques for using NAS in FL.

4.2 Data Analysis

We employ a qualitative research approach, specifically a systematic literature review. In our review, we develop a conceptual model that links violated centralised NAS assumptions to resulting challenges for using NAS in FL, and the techniques used to overcome these challenges as follows:

1. **Challenges arising from Assumption Discrepancies:** We first create a list of challenges that arise when using NAS in FL as a result of the assumption discrepancies between the centralised setting, for which NAS methods tend to get developed, and the FL setting. For each item in our set of relevant literature, we perform open coding to identify challenges, whereby we focus on the motivation of the work with respect to overcoming challenges of NAS in FL in particular. We do not code challenges which occur for reasons other than the aforementioned assumption discrepancies.
2. **Technique Conceptualisation:** Next, we extract techniques for using NAS in FL from the literature in two steps.
 - 2.1 Then, we iteratively refine and merge the unrefined techniques by merging unrefined techniques with conceptually highly similar mechanisms into a single representative technique. Thus, we obtain a coherent set of mutually exclusive and collectively exhaustive techniques for using NAS in FL, whereby a single technique can be a synthesis of techniques from multiple source items of literature.
3. **Techniques for Overcoming Challenges:** Finally, we use axial coding to extract the effect of a technique on overcoming a particular challenge from the items of literature used to synthesise a particular technique.

5 Structure

1. **Introduction** (3 pages)
2. **Background** (6 pages)
 - 2.1 Neural Architecture Search (2.5 pages)
 - 2.2 Federated Learning (2.5 pages)
 - 2.3 NAS in FL (1 page)
3. **Method** (5 pages)
 - 3.1 Reviewed Literature (2 pages)
 - 3.2 Methodology (3 pages)
4. **Challenges of NAS in FL** (10 pages):
 - 4.1 Assumption Discrepancies between centralised NAS and NAS in FL (3 pages)
 - 4.2 Challenges Resulting from Assumption Discrepancies (4 pages)
 - 4.3 Influence of FL system characteristics on Challenges (3 pages)
5. **Techniques for using NAS in FL** (25 pages)
 - 5.1 Technique 1 (1 page)
 - 5.2 Technique 2 (1 page)
 - 5.3 ...
 - 5.4 Technique 20 (1 page)
 - 5.5 Overview (5 pages)
6. **Discussion** (2 pages)
 - 6.1 Principal Findings (1 page)
 - 6.2 Contributions to Practice and Research (0.5 pages)
 - 6.3 Limitations and Future Research (0.5 pages)
7. **Conclusion** (1 page)

52 pages in total.

6 Expected Results

We expect to produce the following:

1. A consolidated catalogue of challenges that arise when using NAS in FL.
2. Classification of the influence of FL system characteristics on the relevance of a challenge.
3. A catalogue of techniques for using NAS in FL with an in-depth discussion of the challenges they overcome.
4. A tabular overview of techniques for using NAS in FL that shows which challenges they overcome at a glance.

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